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Studierichting: Informatica
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$$\int \coth^2 x \cosh x dx$$

$$= \int \frac{\cosh^2 x}{\sinh^2 x} \cosh x dx$$

$$= \int \frac{1 + \sinh^2 x}{\sinh^2 x} \cosh x dx$$

$$= \int \frac{\cosh x}{\sinh^2 x} dx + \int \cosh x dx$$

$$= \int \frac{\cosh x}{\sinh^2 x} dx + \sinh x + C$$

$$u = \sinh x$$

$$du = \cosh x dx$$

$$= \int \frac{\cosh x}{u^2 \cosh x} du + \sinh x + C$$

$$= \int \frac{du}{u^2} + \sinh x + C$$

$$= -\frac{1}{u} + \sinh x + C$$

$$= -\frac{1}{\sinh x} + \sinh x + C$$

References